

E-Commerce Order Management System (OMS)

Document Version: 1.0

Classification: Internal / Architecture

Scope: Core Enterprise OMS

1. Project Overview & Business Objectives

The **Order Management System (OMS)** serves as the centralized engine coordinating commerce across all channels (Web, Mobile, B2B, and 3rd-party marketplaces). It unifies order processing, payment settlement, dynamic distributed order routing (DOM), inventory allocation, and reverse logistics into a resilient workflow pipeline.

Key Business Drivers:

- **Cycle Time Reduction:** Minimize order-to-delivery lead time through end-to-end automated processing and routing.
- **Inventory Accuracy:** Prevent overselling, out-of-stock discrepancies, and high carrying costs across multiple warehouses and stores.
- **Operational Efficiency:** Reduce customer support inquiries with proactive order tracking and self-service RMA workflows.
- **Fulfillment Optimization:** Implement rule-based routing to select optimal fulfillment nodes based on shipping cost, proximity, and capacity.

2. User Roles & Stakeholder Profiles

Role	Core Responsibilities	Key Needs & Expectations
Customer	Places orders, completes payments, monitors delivery status, requests order changes, and initiates returns.	Real-time tracking, immediate confirmation, self-service cancellation and returns.
Store / Order Admin	Monitors pipeline flow, manages order holds, resolves payment/inventory exceptions, and audits fulfillment velocity.	Unified operations dashboard, bulk action workflows, granular status filters.
Warehouse Operator	Executes wave picking, pack-station verification, shipping label generation, and carrier handoffs.	Barcode scanner integration, optimized pick paths, bulk manifest generation.
Customer Support (CSR)	Handles customer disputes, performs address modifications, issues manual discounts, and triggers refunds.	360° order timeline, audit trails, streamlined one-click re-routing and refund controls.

3. Functional Requirements (FR)

FR 1: Multi-Channel Order Capture & Ingestion

- **Channel Ingestion:** Ingest orders in real time via REST and GraphQL APIs from storefronts, mobile apps, and marketplace aggregators (Amazon, eBay, TikTok Shop).
- **Validation Engine:** Perform automated address validation (AVS) against postal registries, compute localized sales tax (e.g., via TaxJar or Avalara), and format contact information.
- **Deduplication & Auditing:** Identify and block idempotent duplicate order requests; record raw ingest payloads for compliance and dispute resolution.

FR 2: Payment Processing & Risk Screening

- **Gateway Integration:** Support pre-authorization, multi-tender capture, manual capture, voids, and partial refunds across gateways (Stripe, PayPal, Adyen).
- **Fraud & Risk Scoring:** Automatically score orders via integrated fraud engines (e.g., Signifyd, Sift, Radar); automatically mark flagged orders as Under Review.
- **Payment Lifecycle Audit:** Maintain a tamper-evident audit record of every authorization code, transaction reference, and state transition.

FR 3: Real-Time Inventory & Distributed Routing (DOM)

- **Soft Reservation:** Reserve inventory temporarily during active checkout sessions with a configurable expiration timeout (e.g., 15 minutes).
- **Hard Allocation:** Permanently deduct available stock and commit quantities upon successful order authorization.
- **Routing Engine:** Dynamically evaluate fulfillment rules: proximity to customer address, warehouse on-hand stock, warehouse operational capacity, and shipping cut-off windows to assign single or split shipments.

FR 4: Warehouse Fulfillment & Carrier Logistics

- **Workflow Transitions:** Transition order line items across distinct states: Created → Routed → Picking → Packed → Shipped → Delivered.
- **Document Generation:** Automatically render digital/printable wave pick lists, packing slips, and carrier-compliant shipping labels (FedEx, UPS, DHL, USPS).
- **Carrier Sync:** Receive automated webhook and EDI tracking updates from carriers, compute estimated delivery dates (EDD), and trigger proactive customer notifications.

FR 5: Reverse Logistics & RMA Management

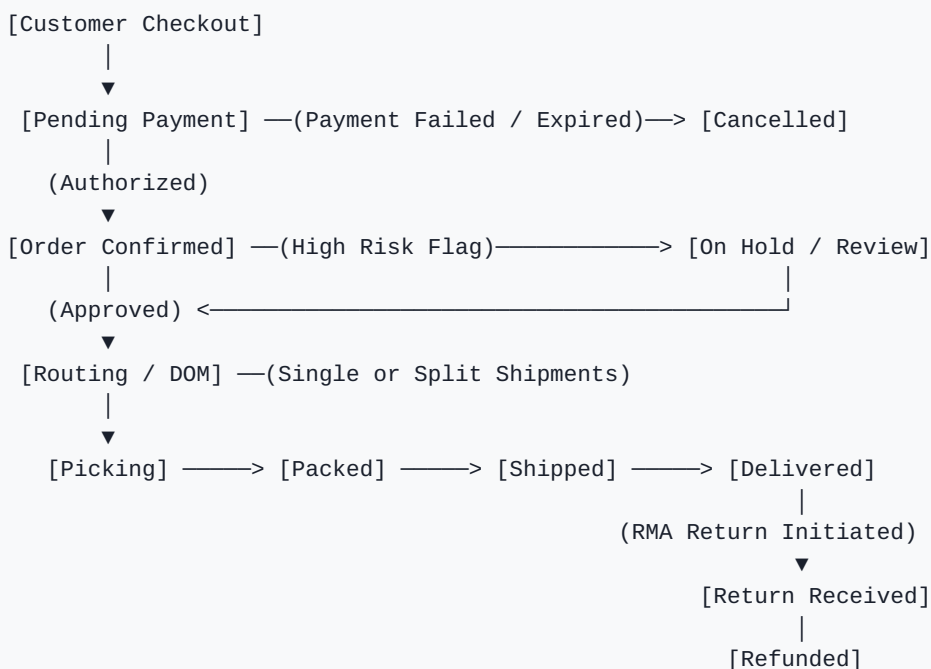
- **Self-Service Return Requests:** Provide customer-facing portals to submit returns within allowed policy windows (e.g., 30 days from delivery).
- **RMA Authorization:** Auto-generate Return Merchandise Authorization (RMA) barcodes and prepaid return shipping slips.
- **Inspection & Disposition:** Allow warehouse triage upon receipt: return acceptable goods to active inventory ledger; mark damaged items for scrap/salvage.
- **Automated Refunds:** Issue partial or full original-tender refunds immediately upon verified return receipt.

4. Non-Functional Requirements (NFR)

Category	Requirement Benchmark
Performance & Latency	95% of synchronous API requests must respond within < 250 ms. The distributed routing engine must process ≥ 5,000 orders/minute under standard load.
High Availability	Target 99.95% system uptime (≤ 4.38 hours downtime/year). Recovery Point Objective (RPO) = 0; Recovery Time Objective (RTO) < 15 minutes.
Scalability	Horizontal auto-scaling to seamlessly handle 10x traffic surges during high-volume sales events (e.g., Black Friday / Cyber Week).
Security & Compliance	PCI-DSS Level 1 compliance; zero unencrypted cardholder data stored. End-to-end encryption with TLS 1.3 in transit and AES-256 at rest. RBAC with MFA for administrative privileges.
Data Auditability	Append-only, immutable event logs for every state transition, financial transaction, and administrative override.

5. Order State Machine & Lifecycle

The state diagram below illustrates the deterministic lifecycle transitions enforced by the OMS execution engine:



6. Core Data Model & Entities

- **Customer** : customer_id, full_name, email, phone, billing_address_id, shipping_address_id, created_at

- **Order** : order_id, order_number, customer_id, channel, status, currency, subtotal, tax_amount, shipping_fee, discount_total, grand_total, created_at
- **OrderItem** : item_id, order_id, sku, product_name, quantity, unit_price, tax_code, fulfillment_status, warehouse_id
- **FulfillmentNode** : node_id, facility_name, facility_type (DC/Store), postal_code, priority_score, capacity_limit, is_active
- **InventoryLedger** : ledger_id, sku, node_id, on_hand, reserved_soft, reserved_hard, safety_stock, updated_at
- **Shipment** : shipment_id, order_id, node_id, carrier_name, tracking_number, label_url, dispatch_date, delivery_date, status
- **ReturnRequest (RMA)** : rma_id, order_id, return_reason, return_status, refund_amount, tracking_number, inspected_at